GIANTS : Generative Insight Anticipation from Scientific Literature

Joy He-Yueya[*1], **Anikait Singh**[*1], **Ge Gao**[1], **Michael Y. Li**[1], **Sherry Yang**[2], **Chelsea Finn**[1], **Emma Brunskill**[1], **Noah D. Goodman**[1]
[1]Stanford University, [2]New York University, [*]Equal contribution

Scientific breakthroughs often emerge from synthesizing prior ideas into novel contributions. While language models (LMs) show promise in scientific discovery, their ability to emulate this targeted, evolutionary synthesis remains largely unexplored. We introduce insight anticipation, a novel literature-grounded generation task that moves beyond open-ended brainstorming by challenging models to predict a downstream paper's core contribution directly from the summaries of its foundational parent papers. To evaluate this capability, we develop GiantsBench, a large-scale benchmark comprising $17k$ tuples of parent and downstream papers across eight scientific domains. We assess performance using an LM judge to measure the similarity between generated and ground-truth insights, demonstrating a positive correlation with human expert evaluations. Finally, we present `GIANTS-4B`, an LM trained via reinforcement learning (RL) to optimize insight anticipation using these similarity scores as a proxy reward. Despite its smaller open-source architecture, `GIANTS-4B` significantly outperforms proprietary baselines, achieving a 34% relative increase in similarity score compared to `gemini-3-pro`. We release our code, dataset, and model to accelerate future research in automated scientific discovery.

"If I have seen further than others, it is only because
I was standing on the shoulders of giants"
— Sir Isaac Newton

1. Introduction

Language Models (LMs) are maturing into powerful tools for scientific discovery [7]. Early milestones are striking, from virtual LM teams designing SARS-CoV-2 nanobody binders [21] to models proposing NLP research directions that human experts rate as genuinely novel [14]. However, these successes rely heavily on prompting existing frontier models that are extensively pre-trained on massive corpora of text. In contrast, human scientists achieve breakthroughs through a far more efficient process: by synthesizing profound insights from sparse and disparate sources. Emulating this human-like capability of conjecturing is not just a theoretical ideal; it is a practical necessity as LMs struggle to reliably generate ideas of true impact and value [15], often due to the lack of diversity and feasibility [23].

To bridge this gap, we propose a new paradigm: *insight anticipation*. Rather than focusing on broad data assimilation, this setting tasks models with predicting the natural evolutionary trajectory of specific, existing research. This framing adopts the classical view of scientific progress as '*standing on the shoulders of giants,*' where new contributions emerge by building incrementally upon prior foundations. Under this view, a paper's value is defined by its relative advancement over the works it builds upon. Consequently, the challenge of automated discovery decomposes into two distinct subproblems: (1) *parent selection*, or identifying which prior works to build upon, and (2) *insight generation*, or synthesizing those works into a novel contribution.

Corresponding author(s): heyueya@stanford.edu, anikait@stanford.edu.
Project Website: https://giants-insights.github.io/

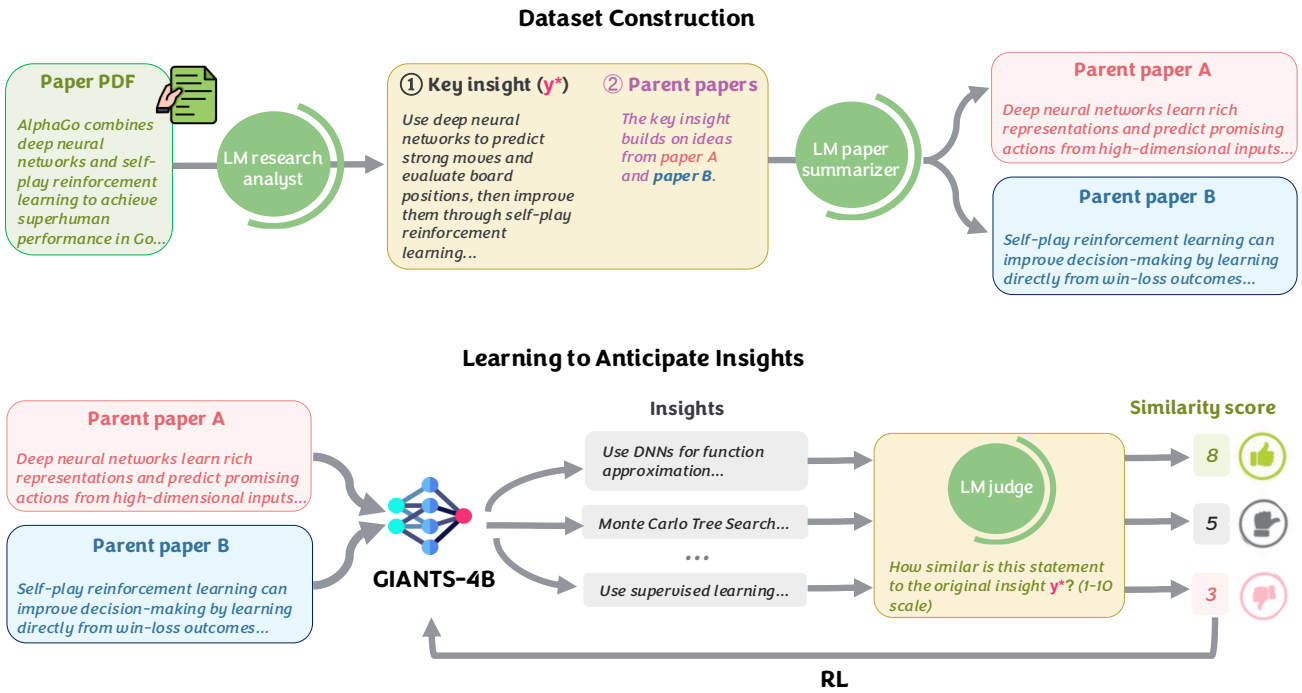

Figure 1: Overview of GiantsBench and GIANTS-4B. *(top)* For dataset construction, we use an LM as a research analyst to process each paper PDF to identify two parent papers whose ideas are synergistically combined to produce the paper's key insight, which is extracted as the ground-truth target y^*. A second LM summarizes each parent paper. *(bottom)* Given the two parent summaries, the model generates candidate insights, which an LM judge scores by similarity to y^* on a 1–10 scale. These scores serve as a proxy reward for RL training, teaching the model to anticipate insights that more closely match real downstream papers.

This motivates a fundamental feasibility question: if provided with the relevant lineage of background papers, can a model effectively predict the next logical leap? In this work, we focus exclusively on the generation phase. By assuming the "parent" papers are provided by an oracle literature selection criterion, we test a model's capacity to transform these foundations into a downstream paper's primary insight. While identifying optimal parent papers remains a complex challenge of scientific attribution, isolating the generation phase allows us to rigorously evaluate the creative potential of targeted synthesis, leaving retrieval strategies for future work.

To evaluate insight anticipation models, we develop GiantsBench, a large-scale benchmark designed to evaluate the ability of models to synthesize two prior papers and derive the core insight of a downstream paper that builds on them (Figure 1, top). Unlike previous benchmarks that evaluate the ability of models to answer literature synthesis questions by identifying relevant papers and generating long-form responses with citations [2], GiantsBench assesses whether a model can combine two parent papers synergistically to derive insights that lead to a subsequent paper. GiantsBench comprises $17k$ input pairs of parent paper summaries and their corresponding downstream research insights, drawn from Computer Science, Economics, Electrical Engineering, Mathematics, Physics, Quantitative Biology, Quantitative Finance, and Statistics. We also introduce an automatic evaluation method that uses an LM judge to give a **similarity score** comparing the model-generated insight to the ground-truth insight. We conducted expert evaluations of the LM judge and found that LM scores and human scores are positively correlated (Spearman's $\rho = 0.761$, $p < 0.001$).

Furthermore, we introduce GIANTS-4B, a language model trained to generate the core insight of a future scientific paper given a set of prior papers via reinforcement learning (RL). By fine-tuning with GRPO [13] to maximize the similarity scores on the research insights, the model learns to reason about connections between the input papers and generate insights that are more similar to those from real downstream papers (Figure 1, bottom). This outperforms simply distilling the expert insight (even with reasoning traces).

We evaluate proprietary and open LMs (e.g., gemini-2.5-pro, gemini-3-pro, Qwen3-4B) on GiantsBench. Qwen3-4B performs similarly as gemini-2.5-pro and gemini-3-pro, despite being a smaller open model, which shows that LMs are not simply getting better at insight anticipation via scaling. In contrast, our GIANTS-4B, which uses RL with similarity scores from the LM judge as a proxy reward, significantly improves insight anticipation.

We summarize our main contributions as follows:

- **Novel paradigm for automated discovery**: We introduce the task of *insight anticipation*, which isolates the creative synthesis phase of scientific progress by evaluating how effective a model is at predicting a downstream paper's core insight conditioned on foundational parent papers.
- **GiantsBench and evaluation metric**: We source a dataset of 17k tuples of parent and downstream papers from arXiv across eight domains and have an LM-based auto-evaluator to score the similarity between generated and ground-truth insights.
- **GIANTS-4B**: We present a novel approach for training a model for insight anticipation based on semantic similarity and RL, outperforming a range of proprietary and open LMs such as gemini-3-pro.

2. Defining and Instantiating a Benchmark for Insight Anticipation

Task Definition. We introduce the setting of insight anticipation: given an input context with two paper summaries $x = (x_A, x_B)$ where x_A is a summary of paper A, and x_B is a summary of paper B, the goal is to generate the key insight of a downstream paper that builds on A and B. We denote this ground truth downstream insight statement as y^*. y^* is an oracle insight that emerges only when both papers are considered together. This task requires the model to consider how A and B might jointly influence a downstream paper and generate a research insight/idea $\hat{y}$ that is semantically similar to y^*. This setup intentionally restricts the number of parents to two (as more papers lead to context management constraints for training and inference), making the task a conservative lower bound on the full problem. We do not address parent discovery, focusing instead on whether meaningful insight synthesis is possible given fixed inputs. Figure 10 shows the insight generation prompt we use for training or evaluating all models.

Dataset Construction. We create a dataset $\{((x_A, x_B)_i, y_i^*)\}_{i=1}^N$ as follows. We collect 17,839 papers from arXiv that are published between May 23rd, 2007, and January 23rd, 2026, according to their latest update date on arXiv. Since arXiv is not peer reviewed, the raw corpus may contain noisy or non-substantive documents. To mitigate this, we retain only papers with at least two citations.[1] We download these papers as a PDF. For each paper, we prompt gemini-2.5-flash to identify two prior

[1]We use citation counts from Semantic Scholar as a proxy for paper quality.

papers that this paper explicitly cites and builds upon by combining their ideas in a synergistic way. We also ask `gemini-2.5-flash` to explain the synergy (see the full prompt in Figure 11). We then download the two parent papers as a PDF and use their content as input context x. Ideally, we would like to take the full paper content as input to the model and ask it to generate a downstream insight. However, given the context-length limitations of existing LMs as well as the high inference cost, we instead opt to prompt `gemini-2.5-flash` to summarize the paper. We then use paper summaries as the input. We use the following prompt to summarize each paper into key insights and contributions: "*Summarize the document, clearly describing the method used and highlighting the key insights or findings. Provide sufficient detail so that the approach and main contributions are fully understood.*" For y^*, we use the synergy explanation that is part of the output to the parent-identification prompt in Figure 11). We cannot use the raw explanation because it refers to a future downstream paper and talks about the insight in the context of the two parent papers. Instead, we would like the model to generate a standalone insight statement directly from the two parent papers alone. Therefore, we prompt `gemini-3-pro` to rewrite the insight and explanation without reference to the downstream paper (see prompt in Appendix C.4). After constructing all (x, y^*) pairs, for each unique pair of parent papers $x = (x_A, x_B)$, we keep the insight y^* from the most cited downstream paper in order to prioritize generating more impactful insights.

Temporal Hold-Out Evaluation. To more rigorously assess generalization, we conduct all evaluations on a *future held-out test set* consisting of downstream papers published after the training cutoff date. We split the dataset by the publication date of the downstream paper in each (x, y^*) pair. Papers published before July 1st, 2023 are used for training. To study the ability of models to generalize to new domains, we restrict the training set to papers tagged with cs.CL (Computation and Language), yielding $N = 10{,}335$ training examples. For evaluation, we consider papers published after July 1, 2023 and randomly sample up to 600 papers from each domain (see the category taxonomy in Appendix A), resulting in a test set of $7{,}504$ papers. No papers from this test set are seen during training, though some of them may share at most 1 parent paper with the papers in the training set. We also conduct evaluations on a subset of the test set that does not have any overlapping parents with those in the training set. We call the subset **Test-unseen-parents** ($N = 5{,}294$).

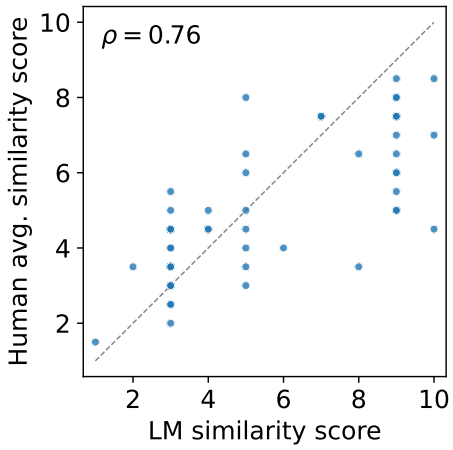

Figure 2: Human experts align with LM judge. LM similarity scores are positively correlated with human similarity scores (Spearman $\rho = 0.761$, $p < 0.001$, $n = 60$).

Evaluation Metric. For each generation, we prompt an LM judge to give a **similarity score** ranging from 1 to 10 (higher is better) comparing the model-generated insight to the ground-truth insight y^*. We use `gemini-3-pro` as the judge model (see Figure 12 for the prompt). To assess the reliability of the LM-based evaluation, we conduct a human evaluation study on 30 pairs of insights generated by `Qwen3-4B` and `GIANTS-4B` ($n = 60$). Two human annotators, who are PhD students in Computer Science, independently rate the similarity between model-generated insights and ground-truth insights using the same rating scale as the LM judge. The LM's scores show a statistically significant positive correlation with the average scores across human annotators (Figure 2). In particular, we observe a

Spearman rank correlation of $\rho = 0.761$ ($p < 0.001$). More details can be found in Appendix E.

Takeaways of Insight Anticipation Instantiation

We introduce the *insight anticipation* task, which challenges models to generate a future research insight by synthesizing the summaries of two parent papers. To evaluate this, we construct a dataset of over 17k arXiv paper triplets with LLM-extracted ground truths, utilizing a strict temporal/cross-domain hold-out split and a human-validated LLM judge for robust scoring.

3. Training an Insight Anticipator

We explore two primary training paradigms to optimize `GIANTS-4B` for insight anticipation: (1) distillation via supervised fine-tuning (SFT), leveraging ground-truth insights and rationalization from target papers, and (2) reinforcement learning (RL) for insight anticipation via similarity optimization. Across all experiments, we adopt `Qwen3-4B` as our base model. Next, we will detail the specific formulations of these training methodologies.

3.1. Ground Truth Insight Distillation via Supervised Fine-Tuning (SFT)

The objective of our SFT pipeline is to adapt the base language model to synthesize a cohesive, derivative insight given the textual context of two source papers. We investigate two distinct SFT strategies to achieve this. In our standard SFT approach, the model is directly fine-tuned to map the input context (summaries of parent papers), x, to the target ground-truth insight, y^* (derived from the target paper). This process optimizes the standard negative log-likelihood loss for autoregressive LMs over the paired examples (x, y^*).

To bridge the logical gap between the source papers and the final insight, we also evaluate an SFT strategy enhanced with chain-of-thought reasoning [26], which we denote as *SFT-think*. In this setup, we introduce an intermediate synthetic rationalization step, z. To generate a supervision target, we prompt a high-capacity teacher model (`gemini-3-pro`) to generate a detailed chain-of-thought that logically deduces the ground-truth insight conditioned on the parent paper summaries (as detailed in Section 2). The base model is then trained to sequentially predict the rationale followed by the final insight, utilizing the augmented tuples (x, z, y^*). This approach explicitly encourages the model to internalize the step-by-step inferential process rather than merely memorizing the final output mapping. This follows distillation approaches in reasoning as seen in works such as OpenThoughts [6] and s1 [9].

3.2. Reinforcement Learning for Insight Anticipation via Similarity Optimization

So far, we have explored distilling insights via supervised fine-tuning (SFT). However, in complex tasks such as scientific discovery, the target paper's insights can be difficult to clone directly. This difficulty often arises when the target insight has low likelihood under the policy, or when the model lacks the capacity to adequately capture the distribution of the parent insight. Can we instead consider an alternative approach that relies on reinforcement learning to recover the performance of the target paper's insights?

To address this, we employ an alternative approach: using the target paper's insights y^* to define a proxy reward via a semantic similarity metric. Formally, given an insight from the target paper y^*, we define

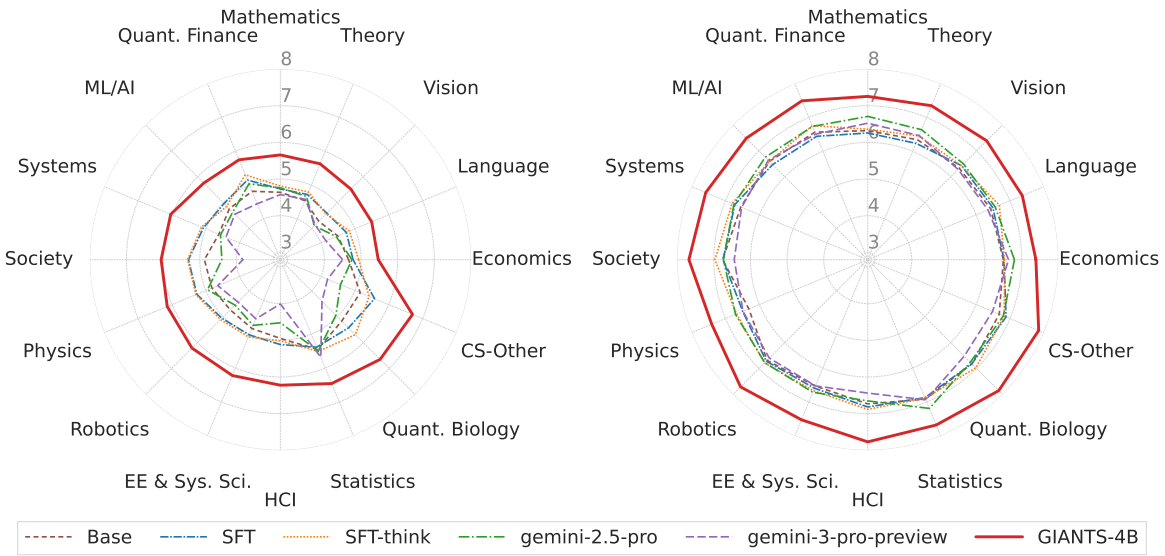

Figure 3: Insight anticipation for arXiv domains. `GIANTS-4B` has a higher similarity score to the target insight, outperforming SFT, SFT-Think, and the base model (`Qwen3-4B`) for two different judge models. The relative ordering of similarity scores for each domain, judged by `gemini-3-pro` *(left)* matches the ordering of similarity scores judged by `Qwen3-14B` *(right)*.

the proxy reward for a predicted insight $\hat{y}$ as:

$$r_{\text{sim}}(\hat{y}) = \textit{similarity}(\hat{y}, y^*) \tag{1}$$

where *similarity* measures the semantic equivalence between the generated insight and target insights measured using an LM-as-a-Judge [5], matching the evaluation criterion in Section 2. Furthermore, for further robustness and reducing order bias, we supply the LM with insights twice ($\hat{y}$ first, and y^* first), taking the average over the individual similarity scores to compute the proxy reward. This proxy reward is then optimized via RL to recover the behavior of the ground-truth insight of the child paper conditioned on the two prespecified parent papers.

More concretely, we train `GIANTS-4B` using Group Relative Policy Optimization (GRPO) [13], utilizing an LM as a proxy reward function. Specifically, for each input context x, we sample a group of $G = 8$ candidate insights from the active policy. An LM judge evaluates these candidates and assigns scalar rewards based on a predefined similarity score metric (illustrated in Figure 1, bottom). GRPO is well-suited for this setup because it optimizes the policy relative to the sampled group, thereby removing the need to train and maintain a separate, memory-intensive value function model.

Crucially, to mitigate the risk of reward hacking and ensure a rigorous evaluation, we enforce a strict separation between the training and testing judges. We deploy `gemini-2.5-flash` as the active reward model during GRPO training, while reserving the independent `gemini-3-pro` model exclusively for the final evaluation phase. This decoupling guarantees a more objective, unbiased assessment of the model's true generalization capabilities. We additionally evaluate with other LM judges (e.g., `Qwen3-14B`) to showcase robustness across model families.

Takeaways of Training an Insight Anticipator

We optimize our insight anticipator using reinforcement learning to maximize semantic similarity to target insights. We strictly separate the LLM judges used for training and final evaluation.

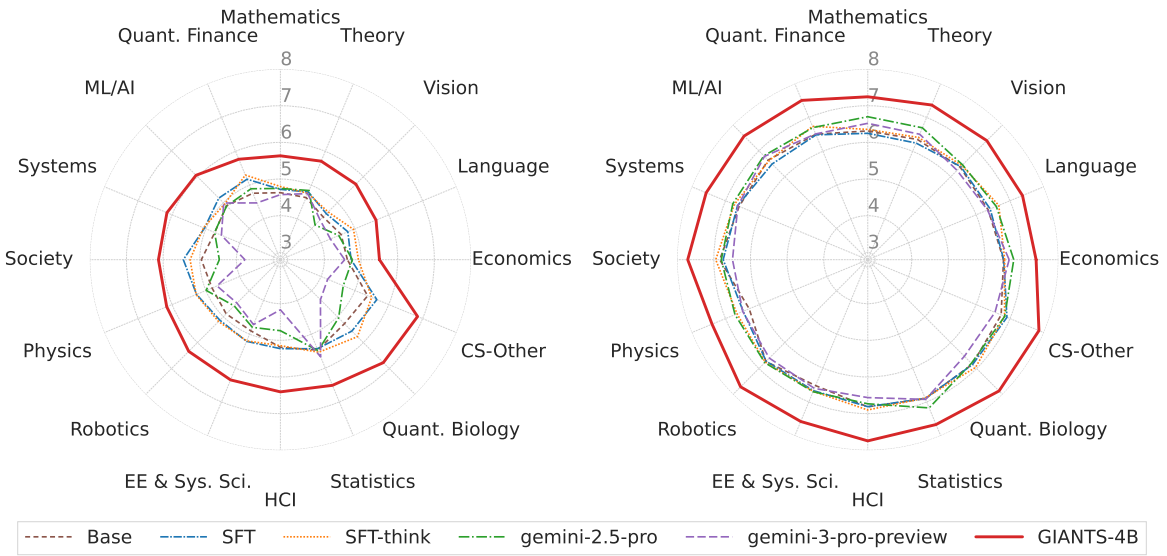

Figure 4: Similarity scores on *Test-unseen-parents*. This is the subset of GiantsBench test set with only unseen parent papers, where GIANTS-4B outperforms the base model and SFT both when judged by gemini-3-pro *(left)* and Qwen3-14B *(right)*.

4. Experimental Evaluation on GiantsBench

The objective of our experimental evaluation is to assess the efficacy of GIANTS-4B in generating novel and plausible scientific insights through the synthesis and expansion of existing literature. We perform a comparative analysis of our approach against both state-of-the-art proprietary models and high-performing open-weight baselines using the GiantsBench dataset. Specifically, our experiments are designed to address the following research questions: (1) Can current frontier and open-source language models natively generate high-quality scientific insights? (2) Does directly optimizing for insight similarity using reinforcement learning (GIANTS-4B) improve idea synthesis compared to standard imitation learning? and (3) Can a model trained to synthesize insights in a single domain generalize its capabilities to unseen domains?

1) Insights generation is a challenging task even for large proprietary models. Despite their success across diverse NLP benchmarks, current frontier models exhibit significant limitations in native idea synthesis. For instance, Qwen3-4B achieves an average similarity score of only 4.75 (Figure 3). Based on our LLM-based evaluation rubric (Figure 12), a score in this range indicates that while generated insights may align topically with the ground truth, they fail to capture the core scientific contribution or technical nuance. Notably, significantly larger proprietary models, such as gemini-2.5-pro and gemini-3-pro, perform similarly to the smaller Qwen3-4B baseline. This finding suggests that scientific ideation capabilities do not scale linearly with model size alone, reinforcing the necessity for specialized training paradigms like GIANTS-4B.

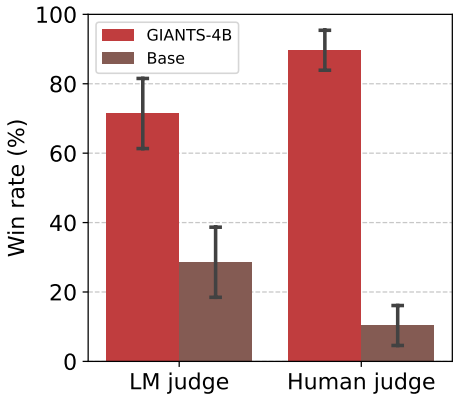

Figure 5: Win rates for GIANTS-4B GIANTS-4B achieves a higher win rate for similarity than the base model for both the LM and Human judge.

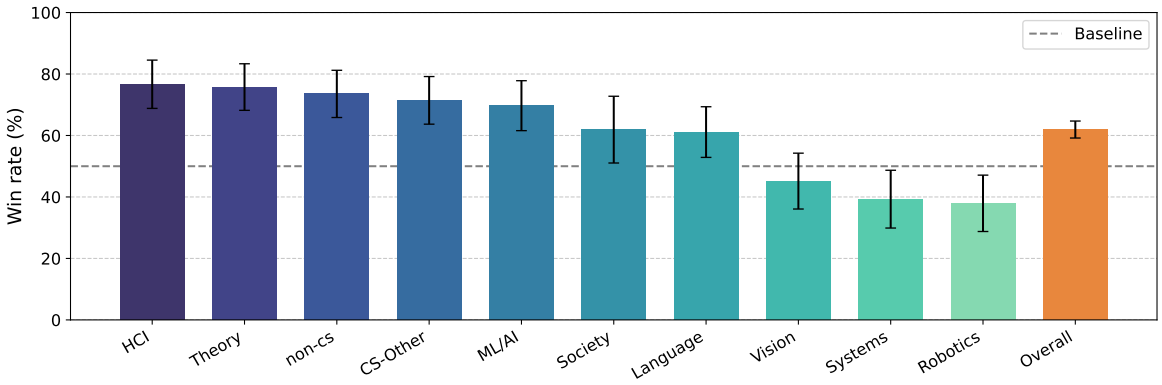

Figure 7: GIANTS-4B achieves a 62% overall win rate against the Base model using SciJudge-30B. We evaluate GIANTS-4B utilizing the judge framework from Tong et al. [22], originally optimized for citation count prediction. While performance varies by domain, reaching up to a 76% win rate in HCI but dropping to 37% in robotics, GIANTS-4B maintains a robust overall win rate of 62%.

2) Directly optimizing similarity scores significantly improves insight generation Our primary finding is that GIANTS-4B achieves the highest performance among all tested methods. By directly optimizing for similarity scores via reinforcement learning (RL), GIANTS-4B yields a relative improvement of 28% on the full test set (Figure 3) and 34% on the subset containing only unseen parent papers (Figure 4) over gemini-3-pro. In contrast, standard supervised fine-tuning (SFT) on ground-truth insights actually degrades performance, while the SFT-think variant only achieves parity with the base model. These results suggest that standard imitation learning is largely ineffective for this synthesis task; however, our RL framework successfully aligns the model's capabilities with high-quality human insights. This performance trend remains robust as we increase the number of samples to optimize the similarity score via inference-time scaling, as illustrated in Figure 8.

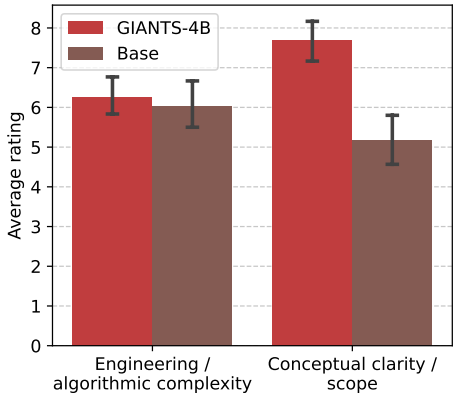

Figure 6: Feasibility Human Eval Humans gauge the feasibility of a generated insight in 2 axes: (1) engineering/algorithmic complexity & (2) conceptual clarity/scope, finding GIANTS-4B generates insights with similar algorithmic complexity and better clarity.

To further validate these gains, we employed the judge framework from Tong et al. [22] as an independent third-party evaluation of paper quality. In this setup, the judge model is fine-tuned to predict which of two papers is more likely to receive higher citations. We found that optimizing our similarity-based metric allows GIANTS-4B to achieve a 62% win rate over the base model across $n = 500$ insights across CS disciplines (Figure 7). This evaluation accounts for potential biases, as in our similarity scoring, by ensuring order invariance and by filtering out inconsistent judgments.

We supplemented this automated evaluation with two independent preliminary human studies to assess the real-world impact of our method. First, we measured the win rate of GIANTS-4B against the base model specifically regarding alignment with the downstream paper's actual insights (Figure 5). In this head-to-head comparison, GIANTS-4B achieved a 71.4% win rate according to an LM judge and a 89.7% win rate according to human evaluators. Finally, we assessed the feasibility of the generated insights across two axes: algorithmic complexity and conceptual clarity. Our analysis indicates that

while GIANTS-4B produces insights of similar algorithmic complexity to the base model, it significantly improves conceptual clarity, making the generated ideas more interpretable and actionable. Further details regarding the human study methodology are provided in Appendix E.

3) GIANTS-4B zero-shot generalizes to new domains and distribution of parent papers. Crucially, the synthesis capabilities acquired by GIANTS-4B are not restricted to its training data. Although trained exclusively on papers from the Language domain (CS.CL), we observed consistent performance gains across all evaluated disciplines as seen in Figure 3, including Physics, Quantitative Biology, and Economics. This suggests that the model learns a generalizable mechanism for combining disparate ideas rather than memorizing domain-specific heuristics. Furthermore, Figure 3 and 4 show the generalization on fully held out papers that appear temporally after the distribution of papers in the training set.

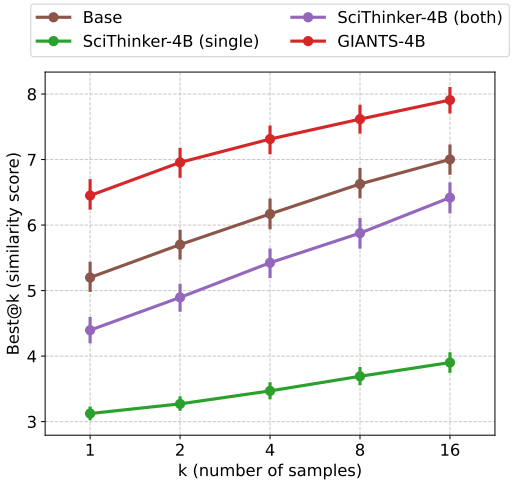

Figure 8: Test Time Inference of GIANTS-4B. GIANTS-4B consistently outperforms the base model and other insight proposer methods such as SciThinker-4B [22], which is optimized for citation count (provided a single or both parent papers).

Evaluation reliability check. To ensure our findings are robust and that similarity scores are not biased toward the specific judge model that was used during training, we conducted a cross-model validation using an independent judge, Qwen3-14B (Figure 3, right). This secondary evaluation corroborated our primary findings, ranking GIANTS-4B as the top-performing model across all metrics, albeit with a slightly more compressed performance margin. We also report results under three additional judge models in Appendix D, with consistent findings across model sizes.

Qualitative Comparison of Models We present qualitative examples to demonstrate that our generated insights meaningfully integrate concepts from prior literature. To study this, we use NeurIPS 2025 award-winning papers as representative high-quality source material and compare the insights produced by GIANTS-4B against those generated by Qwen3-4B.

First, Figure 9 (left) highlights an insight derived from two prior works [8, 24]. While the base model merely summarizes the parent papers without synthesizing a novel perspective, GIANTS-4B proposes a concrete mechanistic connection between the two studies that goes beyond the parent papers. Next, Figure 9 (right) examines an insight based on two different papers [11, 27]. Here, the base model generates an overly ambitious claim, suggesting that gating mechanisms expand reasoning boundaries in ways that RLVR cannot, which is unsupported by the source texts. In contrast, GIANTS-4B remains grounded in the actual findings while still identifying a non-trivial connection: that attention gating may dictate how effectively reinforcement learning concentrates probability mass across useful reasoning trajectories. This comparison underscores a critical distinction between boldness and genuine insightfulness. Whereas the base model's novelty relies on unsupported extrapolation, GIANTS-4B maintains a narrower scope to identify a highly plausible interaction between the foundational works.

Overall, these two examples showcase the qualitative differences between the base model and GIANTS-4B, underscoring the properties of diversity and feasibility that are present in the insights that are generated.

Example 1

Paper 1:
Scaling network depth up to 1024 layers in contrastive reinforcement learning yields 2–50x performance gains and emergent complex behaviors across locomotion and manipulation tasks.

Paper 2:
Representation superposition, where models encode more features than their hidden dimensions, mechanistically explains neural scaling laws in LLMs, with strong superposition producing robust inverse-dimension loss scaling regardless of data distribution.

Base

*The depth scaling in self-supervised reinforcement learning (Paper 1) and the model size scaling in large language models (Paper 2) both rely on a shared principle of **representation superposition**, but their mechanisms diverge based on the nature of the environment. In Paper 1, deeper networks enable **richer contrastive representations** that capture complex environmental topologies (e.g., maze layouts), which aligns with the "strong superposition" regime in Paper 2, where overlapping representations in LLMs allow robust power-law scaling of loss with model size. However, Paper 1's depth scaling achieves this through **synergistic exploration and expressivity**, whereas Paper 2's strong superposition depends on **geometric overlaps** between representation vectors. This suggests that in RL, depth scaling enhances the agent's ability to explore and generalize through **targeted representational capacity allocation**, while in LLMs, strong superposition leverages **heterogeneous representation vectors** to handle diverse data. The critical distinction lies in how these scaling strategies interact with the environment's structure: depth in RL unlocks emergent behaviors via **partial experience stitching**, while LLMs' superposition ensures **robustness across varied data distributions**. Together, this implies that **scaling strategies in different modalities (RL vs. LLMs)** can be unified under the concept of **representation superposition**, but their effectiveness hinges on the specific interplay between model capacity, environmental complexity, and the underlying representation mechanisms.*

GIANTS-4B

*Deep neural networks learn rich representations and predict promising actions from high-dimensional inputs... By synthesizing the *representation superposition* mechanism from Paper 2 (LLMs) with the *depth-driven scaling* from Paper 1 (self-supervised RL), we uncover a novel hypothesis: deep, residual-structured networks in goal-conditioned RL could *explicitly encode geometrically rich, superposed state-action-goal embeddings* that align with the *power-law loss scaling* observed in LLMs. This would mean that increasing network depth in Paper 1's CRL framework not only enhances policy expressivity but also *systematically improves the model's ability to generalize across unseen trajectories* by leveraging the *representation redundancy* and *interference patterns* inherent to superposition. The key implication is that the *contrastive alignment* of embeddings in Paper 1's InfoNCE objective could be interpreted as a *formalization of superposition*, where deeper networks better "stitch" partial experiences into a coherent, high-capacity latent space, mirroring the scaling laws of LLMs.*

Example 2

Paper 1:
RLVR training improves how efficiently LLMs sample correct answers but doesn't expand their underlying reasoning capabilities beyond what the base model already knows, as shown by base models outperforming RLVR models at high pass@k.

Paper 2:
Adding a head-specific sigmoid gate after the scaled dot-product attention output boosts LLM performance, training stability, and long-context extrapolation by introducing beneficial non-linearity and input-dependent sparsity that mitigates attention sinks.

Base

The integration of gating mechanisms (as shown in Paper 2) with the base model's attention dynamics can enhance its reasoning capacity in ways that RLVR (Paper 1) does not achieve. While RLVR primarily sharpens the base model's prior distribution toward high-reward paths without introducing new reasoning patterns, the head-specific sigmoid gate applied after SDPA output (G1) in Paper 2 introduces input-dependent sparsity and non-linearity. This alters the base model's attention dynamics, enabling it to focus on critical reasoning paths more effectively and mitigate issues like "attention sink" or "massive activation." By doing so, the gating mechanism allows the base model to solve problems that require nuanced, multi-step reasoning without relying on RLVR's sampling efficiency gains, suggesting that attentional sparsity and non-linearity can expand the base model's reasoning boundary in a fundamentally different way than reinforcement learning. This implies a new research direction: combining gating-based attention optimization with distillation or other knowledge transfer methods to achieve both efficiency and novel reasoning in LLMs.

GIANTS-4B

*By merging the two studies, we uncover a hidden interplay between how LLMs reason and how their internal attention mechanisms shape that reasoning. When a model's attention is gated to focus on relevant information (as shown in Paper 2), its underlying reasoning paths—measured by metrics like `pass@k` from Paper 1—become more structured and efficient. This gating, for example, might filter out noise or irrelevant tokens, allowing the model to sample fewer but more accurate reasoning steps during reinforcement training. In Paper 1, RLVR's "improvement" is often attributed to refining existing reasoning patterns, but with optimized attention, those patterns could align more closely with true logical structures, making the model's reasoning *appear* more robust even if it's still constrained by the base distribution.*
*The results suggest that the attention layer's dynamics directly influence the "reasoning boundary" discussed in Paper 1. A gate that sparsely prioritizes key information (like query-dependent features) could narrow the set of solvable problems in a way that complements RLVR's focus on sampling efficiency. For instance, a model with better attention might generate fewer but more reliable CoT steps, improving `pass@k` at lower `k` values without expanding the set of solvable problems. This synergy implies that future work could combine RLVR with attentional refinements to not just "narrow" reasoning but *refine* it, enabling models to handle complex tasks with fewer, more targeted attempts. The insight also hints that attention-based mechanisms like gating could act as a bottleneck for reasoning expansion, where RLVR's reinforcement is limited by the model's ability to focus on the right information, not just the right rewards.*

Figure 9: Qualitative comparison of insights derived from NeurIPS 2025 award-winning papers. (Left) GIANTS-4B identifies a more concrete cross-paper mechanism than the base model (Qwen3-4B). (Right) GIANTS-4B produces a grounded, meaningful contribution, avoiding the overly ambitious and infeasible propositions generated by the base model.

Takeaways of Experimental Evaluation

GIANTS-4B significantly outperforms both frontier models and supervised fine-tuning at generating feasible, high-quality scientific insights. Furthermore, despite training exclusively on a single domain, the model successfully zero-shot generalizes its synthesis capabilities across diverse, unseen scientific disciplines and temporally held-out literature.

5. Related Work

AI for Research. Many works have studied how to use LMs for different components of the scientific research pipeline, including literature search [10, 28], idea generation [14, 25], and idea execution [7, 15, 16]. A concurrent work trains models to judge research impact from citation signals, then uses the learned judge as a reward model for idea generation [22]. In contrast, we focus on the problem of synthesizing insights from two parent papers, using similarity to ground-truth downstream insights as a more direct training signal. A recent line of work studies scientific progress as a forecasting problem. Most closely related concurrent work introduces PreScience, a benchmark that decomposes the research process into collaborator prediction, prior work selection, contribution generation, and impact prediction [1]. In their contribution generation task, models generate a future paper's title and abstract conditioned on prior work and other historical context, and are evaluated using an LM-based similarity metric. Our work is complementary but more focused: rather than modeling the full scientific workflow, we isolate

the idea-synthesis problem and ask whether a model can generate the core insight of a downstream paper from a given pair of parent papers. Thus, while PreScience studies broad scientific forecasting, we study a more controlled and direct test of whether models can synthesize a novel downstream insight from specific prior works.

Literature-based Discovery. Literature-Based Discovery (LBD) is a field that involves using computational methods to uncover hidden links between seemingly unrelated bodies of research, aiming to infer novel and interesting knowledge [18, 20]. For instance, [19] noticed that papers on fish oil mentioned it reduces blood viscosity, while others linked high blood viscosity to Raynaud's syndrome. Therefore, [19] hypothesized that fish oil could treat Raynaud's, a prediction later confirmed by clinical research. However, existing LBD approaches struggle to scale to the massive volume of today's literature and often rely on domain-specific knowledge sources. Moreover, they typically output only ranked lists of candidate connections rather than clear, hypothesis-like insights that researchers can act on [3, 4, 12, 17].

6. Discussion, Limitations, Future Work

In this work, we introduced *insight anticipation* as a measurable paradigm for automated scientific discovery. By isolating the insight generation phase, we demonstrated that models can effectively synthesize the core insight of a downstream paper when provided with its foundational parent papers. Our results using GIANTS-4B indicate that the trajectory of scientific intuition is partially predictable, and that optimizing language models via reinforcement learning with similarity-based rewards is a highly effective training strategy for this task. However, this foundational formulation leaves several important limitations and open questions.

One limitation of the work is that we assume that downstream contributions derive from two parent papers due to context constraints, despite research ideas being heavily shaped by broader intellectual contexts. Furthermore, parent identification is imperfect. Citations do not always reflect true conceptual influence, and influential ideas may remain uncited. Finally, by assuming an oracle literature selection criterion, we explicitly decoupled insight generation from parent selection, which may not be feasible for some scientific tasks.

Future research can integrate automated retrieval systems to unify parent selection and synthesis within a single end-to-end framework. Additionally, extending the task to accommodate multi-source lineage and developing evaluation metrics that prioritize conceptual novelty over textual similarity could be interesting. Ultimately, testing these models in active, human-in-the-loop research settings will determine their true potential as catalysts for scientific discovery.

7. Ethics Statement

This research involves training the GIANTS-4B model on a dataset of 17,839 publicly available arXiv preprints to synthesize scientific insights. While automating scientific ideation offers significant potential for discovery, it raises important ethical considerations regarding the proper attribution of foundational ideas, especially since academic citation graphs can be noisy and may not always reflect true conceptual influence. Additionally, there is an inherent risk of language models generating plausible but unverified scientific claims; our framework mitigates this by using reinforcement learning to directly align model outputs with grounded, human-derived insights. Finally, we acknowledge the environmental and computational

costs associated with training models on high-performance GPUs, though utilizing a more efficient 4B parameter architecture helps limit this footprint compared to massive proprietary models.

8. Reproducibility Statement

To ensure the reproducibility of our results, we provide a comprehensive account of our methodology, code, and data. The source code for our models and experiments is available in the supplementary materials and at the following repository: https://github.com/ and our benchmark and model weights can be found in the following huggingface respository: https://huggingface.co/giants2026. Our implementation is built upon the verl framework (https://verl.readthedocs.io/en/latest/). All experimental details, including hyperparameter settings, are documented in Appendix C.5. The computational experiments were conducted on a machine with NVIDIA A100 GPUs, and the required software dependencies are listed in the requirements.txt file within our code repository. The datasets used in our experiments will be made publicly available. We include samples of the dataset in the Appendix along with specific splits and any preprocessing steps applied to the data.

9. Acknowledgments

We thank Shirley Wu, Chenglei Si, Ryan Louie, Rui Li, Aviral Kumar, and Andreas Stuhlmüller and others in the Stanford Coco Labs and Iris Labs for discussions and feedback. This work was supported by the Junglee Corporation Stanford Graduate Fellowship. AS gratefully acknowledges the support of the NSF Graduate Research Fellowship Program. CF was supported by Schmidt Sciences.

References

[1] Anirudh Ajith, Amanpreet Singh, Jay DeYoung, Nadav Kunievsky, Austin C Kozlowski, Oyvind Tafjord, James Evans, Daniel S Weld, Tom Hope, and Doug Downey. Prescience: A benchmark for forecasting scientific contributions. *arXiv preprint arXiv:2602.20459*, 2026.

[2] Akari Asai, Jacqueline He, Rulin Shao, Weijia Shi, Amanpreet Singh, Joseph Chee Chang, Kyle Lo, Luca Soldaini, Sergey Feldman, Mike D'Arcy, et al. Synthesizing scientific literature with retrieval-augmented language models. *Nature*, pages 1–7, 2026.

[3] Tanja Bekhuis. Conceptual biology, hypothesis discovery, and text mining: Swanson's legacy. *Biomedical digital libraries*, 3(1):2, 2006.

[4] Murat C Ganiz, William M Pottenger, and Christopher D Janneck. Recent advances in literature based discovery. *Journal of the American Society for Information Science and Technology, JASIST (Submitted)*, 2005.

[5] Jiawei Gu, Xuhui Jiang, Zhichao Shi, Hexiang Tan, Xuehao Zhai, Chengjin Xu, Wei Li, Yinghan Shen, Shengjie Ma, Honghao Liu, Saizhuo Wang, Kun Zhang, Yuanzhuo Wang, Wen Gao, Lionel Ni, and Jian Guo. A survey on llm-as-a-judge, 2025. URL https://arxiv.org/abs/2411.15594.

[6] Etash Guha, Ryan Marten, Sedrick Keh, Negin Raoof, Georgios Smyrnis, Hritik Bansal, Marianna Nezhurina, Jean Mercat, Trung Vu, Zayne Sprague, Ashima Suvarna, Benjamin Feuer, Liangyu Chen, Zaid Khan, Eric Frankel, Sachin Grover, Caroline Choi, Niklas Muennighoff, Shiye Su, Wanjia Zhao, John Yang, Shreyas Pimpalgaonkar, Kartik Sharma, Charlie Cheng-Jie Ji, Yichuan Deng, Sarah Pratt, Vivek Ramanujan, Jon Saad-Falcon, Jeffrey Li, Achal Dave, Alon Albalak, Kushal Arora, Blake Wulfe, Chinmay Hegde, Greg Durrett, Sewoong Oh, Mohit Bansal, Saadia Gabriel, Aditya Grover, Kai-Wei Chang, Vaishaal Shankar, Aaron Gokaslan, Mike A. Merrill, Tatsunori Hashimoto, Yejin Choi, Jenia Jitsev, Reinhard Heckel, Maheswaran Sathiamoorthy, Alexandros G. Dimakis, and Ludwig Schmidt. Openthoughts: Data recipes for reasoning models, 2025. URL https://arxiv.org/abs/2506.04178.

[7] Andrej Karpathy. autoresearch: Ai agents running research on single-gpu nanochat training automatically. https://github.com/karpathy/autoresearch, 2026.

[8] Yizhou Liu, Ziming Liu, and Jeff Gore. Superposition yields robust neural scaling. *arXiv preprint arXiv:2505.10465*, 2025.

[9] Niklas Muennighoff, Zitong Yang, Weijia Shi, Xiang Lisa Li, Li Fei-Fei, Hannaneh Hajishirzi, Luke Zettlemoyer, Percy Liang, Emmanuel Candès, and Tatsunori Hashimoto. s1: Simple test-time scaling, 2025. URL https://arxiv.org/abs/2501.19393.

[10] OpenAI. Introducing deep research. https://openai.com/index/introducing-deep-research/, 2025. Accessed: 2025-02-02.

[11] Zihan Qiu, Zekun Wang, Bo Zheng, Zeyu Huang, Kaiyue Wen, Songlin Yang, Rui Men, Le Yu, Fei Huang, Suozhi Huang, et al. Gated attention for large language models: Non-linearity, sparsity, and attention-sink-free. *arXiv preprint arXiv:2505.06708*, 2025.

[12] Yakub Sebastian, Eu-Gene Siew, and Sylvester O Orimaye. Emerging approaches in literature-based discovery: techniques and performance review. *The Knowledge Engineering Review*, 32:e12, 2017.

[13] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, YK Li, Yang Wu, et al. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.

[14] Chenglei Si, Diyi Yang, and Tatsunori Hashimoto. Can llms generate novel research ideas? a large-scale human study with 100+ nlp researchers. *arXiv preprint arXiv:2409.04109*, 2024.

[15] Chenglei Si, Tatsunori Hashimoto, and Diyi Yang. The ideation-execution gap: Execution outcomes of llm-generated versus human research ideas. *arXiv preprint arXiv:2506.20803*, 2025.

[16] Chenglei Si, Zitong Yang, Yejin Choi, Emmanuel Candès, Diyi Yang, and Tatsunori Hashimoto. Towards execution-grounded automated ai research. *arXiv preprint arXiv:2601.14525*, 2026.

[17] Neil R Smalheiser. Literature-based discovery: Beyond the abcs. *Journal of the American Society for Information Science and Technology*, 63(2):218–224, 2012.

[18] Don R Swanson. Fish oil, raynaud's syndrome, and undiscovered public knowledge. *Perspectives in biology and medicine*, 30(1):7–18, 1986.

[19] Don R Swanson. Undiscovered public knowledge. *The Library Quarterly*, 56(2):103–118, 1986.

[20] DR Swanson. Literature-based discovery? the very idea. In *Literature-based discovery*, pages 3–11. Springer, 2008.

[21] Kyle Swanson, Wesley Wu, Nash L Bulaong, John E Pak, and James Zou. The virtual lab of ai agents designs new sars-cov-2 nanobodies. *Nature*, pages 1–3, 2025.

[22] Jingqi Tong, Mingzhe Li, Hangcheng Li, Yongzhuo Yang, Yurong Mou, Weijie Ma, Zhiheng Xi, Hongji Chen, Xiaoran Liu, Qinyuan Cheng, et al. Ai can learn scientific taste. *arXiv preprint arXiv:2603.14473*, 2026.

[23] Manya Wadhwa, Tiasa Singha Roy, Harvey Lederman, Junyi Jessy Li, and Greg Durrett. Create: Testing llms for associative creativity, 2026. URL https://arxiv.org/abs/2603.09970.

[24] Kevin Wang, Ishaan Javali, MichaĹ Bortkiewicz, Benjamin Eysenbach, et al. 1000 layer networks for self-supervised rl: Scaling depth can enable new goal-reaching capabilities. *arXiv preprint arXiv:2503.14858*, 2025.

[25] Qingyun Wang, Doug Downey, Heng Ji, and Tom Hope. SciMON: Scientific inspiration machines optimized for novelty. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 279–299, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.18. URL https://aclanthology.org/2024.acl-long.18/.

[26] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models, 2023. URL https://arxiv.org/abs/2201.11903.

[27] Yang Yue, Zhiqi Chen, Rui Lu, Andrew Zhao, Zhaokai Wang, Shiji Song, and Gao Huang. Does reinforcement learning really incentivize reasoning capacity in llms beyond the base model? *arXiv preprint arXiv:2504.13837*, 2025.

[28] Yuxiang Zheng, Dayuan Fu, Xiangkun Hu, Xiaojie Cai, Lyumanshan Ye, Pengrui Lu, and Pengfei Liu. DeepResearcher: Scaling deep research via reinforcement learning in real-world environments. In Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng, editors, *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 414–431, Suzhou, China, November 2025. Association for Computational Linguistics. ISBN 979-8-89176-332-6. doi: 10.18653/v1/2025.emnlp-main.22. URL https://aclanthology.org/2025.emnlp-main.22/.

A. arXiv Paper Domain Classification

We use the arXiv category taxonomy. Since Computer Science has many subfields, we further group the arXiv category labels into broader macro-domains.

Domain	arXiv Labels
Language	cs.CL
ML/AI	cs.LG, cs.AI, cs.NE, cs.MA
Robotics	cs.RO
Vision	cs.CV, cs.GR, cs.MM, cs.SD
Theory	cs.CC, cs.DS, cs.FL, cs.LO, cs.DM, cs.CG, cs.GT, cs.CR, cs.IT
Systems	cs.AR, cs.OS, cs.DC, cs.NI, cs.PF, cs.SY, cs.PL, cs.SE, cs.DB, cs.IR, cs.SI
Society	cs.CY
HCI	cs.HC
CS-Other	cs.ET, cs.GL, cs.OH, cs.DL, cs.NA, cs.MS, cs.CE, cs.SC
Economics	econ
Electrical Engineering & Systems Science (EE & Sys. Sci.)	eess
Mathematics	math
Physics	astro-ph, cond-mat, gr-qc, hep-ex, hep-lat, hep-ph, hep-th, math-ph, nlin, nucl-ex, nucl-th, physics, quant-ph
Quantitative Biology (Quant. Bio.)	q-bio
Quantitative Finance (Quant. Fin.)	q-fin
Statistics	stat

Table 1: Mapping from domains to arXiv category labels.

B. Further Quantitative Analysis

In this section, we provide a more granular, domain-by-domain breakdown of our quantitative results to better understand the performance characteristics of each model across different scientific disciplines.

Table 2 presents the average similarity scores across the entirety of GIANTSBENCH. The results demonstrate that GIANTS-4B achieves consistent, robust improvements over both the base models and the supervised fine-tuning (SFT) baselines. Notably, this performance gap holds steady across highly diverse fields, ranging from largely theoretical domains like Mathematics and Theory to applied disciplines such as Vision, Robotics, and HCI.

To further validate the generalization capabilities of our approach and ensure the model is not merely memorizing training distributions, we separately evaluated performance on a highly constrained subset of the data: **Test-unseen-parents**. This subset exclusively contains test examples where the parent papers were never encountered during the training phase.

The results for this holdout set are detailed in Table 3. Encouragingly, GIANTS-4B maintains its substantial performance gap over the baselines even on unseen literature. This indicates that our model has successfully learned generalized mechanisms for cross-paper insight generation that translate effectively to novel scientific texts.

C. Prompts

This section details the exact prompt templates utilized throughout our pipeline for both generation and evaluation tasks.

Domain	Base	SFT	SFT-think	gemini-2.5-pro	gemini-3-pro	GIANTS-4B
Economics	$4.66_{\pm 0.21}$	$4.78_{\pm 0.20}$	$4.94_{\pm 0.21}$	$4.75_{\pm 0.22}$	$4.50_{\pm 0.20}$	$5.47_{\pm 0.22}$
Language	$4.50_{\pm 0.10}$	$4.75_{\pm 0.10}$	$4.84_{\pm 0.11}$	$4.45_{\pm 0.11}$	$4.14_{\pm 0.10}$	$5.50_{\pm 0.11}$
Vision	$4.28_{\pm 0.10}$	$4.59_{\pm 0.10}$	$4.57_{\pm 0.10}$	$4.11_{\pm 0.10}$	$4.14_{\pm 0.10}$	$5.52_{\pm 0.12}$
Theory	$4.54_{\pm 0.10}$	$4.72_{\pm 0.10}$	$4.80_{\pm 0.11}$	$4.67_{\pm 0.11}$	$4.58_{\pm 0.11}$	$5.63_{\pm 0.11}$
Mathematics	$4.64_{\pm 0.11}$	$4.73_{\pm 0.10}$	$4.81_{\pm 0.10}$	$4.76_{\pm 0.11}$	$4.58_{\pm 0.10}$	$5.66_{\pm 0.11}$
Quant. Fin.	$4.82_{\pm 0.25}$	$5.15_{\pm 0.23}$	$5.31_{\pm 0.25}$	$5.04_{\pm 0.26}$	$4.45_{\pm 0.24}$	$5.75_{\pm 0.26}$
ML/AI	$4.77_{\pm 0.11}$	$4.97_{\pm 0.11}$	$4.87_{\pm 0.11}$	$4.64_{\pm 0.11}$	$4.56_{\pm 0.11}$	$5.76_{\pm 0.12}$
Systems	$4.62_{\pm 0.11}$	$5.07_{\pm 0.11}$	$5.13_{\pm 0.11}$	$4.57_{\pm 0.11}$	$4.41_{\pm 0.11}$	$6.04_{\pm 0.12}$
Society	$4.87_{\pm 0.14}$	$5.31_{\pm 0.14}$	$5.33_{\pm 0.15}$	$4.40_{\pm 0.13}$	$3.82_{\pm 0.11}$	$6.05_{\pm 0.15}$
Physics	$4.82_{\pm 0.11}$	$5.27_{\pm 0.11}$	$5.30_{\pm 0.11}$	$4.95_{\pm 0.11}$	$4.65_{\pm 0.11}$	$6.14_{\pm 0.12}$
Robotics	$4.74_{\pm 0.11}$	$5.06_{\pm 0.11}$	$5.11_{\pm 0.11}$	$4.55_{\pm 0.11}$	$4.43_{\pm 0.11}$	$6.21_{\pm 0.11}$
EE & Sys.	$4.83_{\pm 0.11}$	$5.02_{\pm 0.11}$	$5.08_{\pm 0.11}$	$4.74_{\pm 0.11}$	$4.55_{\pm 0.11}$	$6.22_{\pm 0.12}$
HCI	$4.95_{\pm 0.12}$	$5.12_{\pm 0.11}$	$5.02_{\pm 0.11}$	$4.52_{\pm 0.11}$	$4.01_{\pm 0.10}$	$6.23_{\pm 0.12}$
Statistics	$5.45_{\pm 0.16}$	$5.38_{\pm 0.15}$	$5.51_{\pm 0.15}$	$5.50_{\pm 0.16}$	$5.68_{\pm 0.17}$	$6.46_{\pm 0.16}$
Quant. Bio.	$5.17_{\pm 0.17}$	$5.43_{\pm 0.17}$	$5.69_{\pm 0.17}$	$4.94_{\pm 0.17}$	$4.41_{\pm 0.15}$	$6.65_{\pm 0.17}$
CS-Other	$5.16_{\pm 0.16}$	$5.58_{\pm 0.16}$	$5.42_{\pm 0.16}$	$4.57_{\pm 0.15}$	$4.19_{\pm 0.14}$	$6.70_{\pm 0.16}$
Overall	$4.75_{\pm 0.03}$	$5.01_{\pm 0.03}$	$5.05_{\pm 0.03}$	$4.65_{\pm 0.03}$	$4.43_{\pm 0.03}$	$5.97_{\pm 0.03}$

Table 2: Average similarity scores (mean $\pm$ standard error) per domain on GIANTSBENCH. The judge LM is gemini-3-pro.

Domain	Base	SFT	SFT-think	gemini-2.5-pro	gemini-3-pro	GIANTS-4B
Economics	$4.63_{\pm 0.22}$	$4.73_{\pm 0.21}$	$4.93_{\pm 0.21}$	$4.76_{\pm 0.23}$	$4.55_{\pm 0.21}$	$5.50_{\pm 0.23}$
Language	$4.59_{\pm 0.19}$	$4.79_{\pm 0.19}$	$4.96_{\pm 0.20}$	$4.52_{\pm 0.19}$	$4.28_{\pm 0.18}$	$5.62_{\pm 0.20}$
Vision	$4.45_{\pm 0.14}$	$4.57_{\pm 0.15}$	$4.64_{\pm 0.14}$	$4.14_{\pm 0.14}$	$4.33_{\pm 0.15}$	$5.71_{\pm 0.17}$
Theory	$4.63_{\pm 0.12}$	$4.80_{\pm 0.12}$	$4.76_{\pm 0.12}$	$4.84_{\pm 0.13}$	$4.75_{\pm 0.13}$	$5.71_{\pm 0.13}$
Mathematics	$4.63_{\pm 0.11}$	$4.72_{\pm 0.10}$	$4.80_{\pm 0.10}$	$4.74_{\pm 0.11}$	$4.57_{\pm 0.10}$	$5.63_{\pm 0.11}$
Quant. Fin.	$4.76_{\pm 0.27}$	$5.18_{\pm 0.25}$	$5.30_{\pm 0.26}$	$4.89_{\pm 0.27}$	$4.48_{\pm 0.25}$	$5.77_{\pm 0.27}$
ML/AI	$4.88_{\pm 0.15}$	$5.18_{\pm 0.16}$	$4.95_{\pm 0.15}$	$4.87_{\pm 0.16}$	$4.99_{\pm 0.16}$	$6.06_{\pm 0.16}$
Systems	$4.74_{\pm 0.15}$	$5.04_{\pm 0.15}$	$5.07_{\pm 0.15}$	$4.74_{\pm 0.16}$	$4.54_{\pm 0.15}$	$6.15_{\pm 0.16}$
Society	$4.96_{\pm 0.19}$	$5.45_{\pm 0.19}$	$5.26_{\pm 0.19}$	$4.47_{\pm 0.17}$	$3.77_{\pm 0.14}$	$6.13_{\pm 0.20}$
Physics	$4.82_{\pm 0.11}$	$5.28_{\pm 0.11}$	$5.29_{\pm 0.11}$	$4.99_{\pm 0.12}$	$4.67_{\pm 0.11}$	$6.16_{\pm 0.12}$
Robotics	$4.89_{\pm 0.13}$	$5.14_{\pm 0.12}$	$5.19_{\pm 0.13}$	$4.58_{\pm 0.13}$	$4.48_{\pm 0.13}$	$6.34_{\pm 0.13}$
EE & Sys.	$4.90_{\pm 0.13}$	$5.21_{\pm 0.13}$	$5.20_{\pm 0.12}$	$4.80_{\pm 0.13}$	$4.72_{\pm 0.13}$	$6.35_{\pm 0.14}$
HCI	$5.19_{\pm 0.15}$	$5.23_{\pm 0.14}$	$5.15_{\pm 0.14}$	$4.74_{\pm 0.14}$	$4.17_{\pm 0.12}$	$6.41_{\pm 0.15}$
Statistics	$5.47_{\pm 0.17}$	$5.43_{\pm 0.15}$	$5.53_{\pm 0.16}$	$5.49_{\pm 0.17}$	$5.67_{\pm 0.18}$	$6.51_{\pm 0.16}$
Quant. Bio.	$5.27_{\pm 0.18}$	$5.56_{\pm 0.18}$	$5.77_{\pm 0.18}$	$5.05_{\pm 0.18}$	$4.35_{\pm 0.16}$	$6.77_{\pm 0.18}$
CS-Other	$5.38_{\pm 0.19}$	$5.65_{\pm 0.18}$	$5.50_{\pm 0.18}$	$4.66_{\pm 0.17}$	$4.20_{\pm 0.16}$	$6.85_{\pm 0.18}$
Overall	$4.87_{\pm 0.04}$	$5.10_{\pm 0.04}$	$5.11_{\pm 0.04}$	$4.78_{\pm 0.04}$	$4.57_{\pm 0.04}$	$6.11_{\pm 0.04}$

Table 3: Average similarity scores (mean $\pm$ standard error) per domain on **Test-unseen-parents**, which indicates the subset of GIANTSBENCH test set with only unseen parent papers. The judge LLM is gemini-3-pro.

You are given summaries of two research papers.
<papers>
Paper 1:
{Paper 1 summary}

Paper 2:
{Paper 2 summary}
</papers>

Your task is to generate a novel and non-obvious insight that emerges only when both papers are considered together.

Quality requirements:
1) SPECIFIC & PRECISE: Ground the insight in fine-grained details (methods, mechanisms, results, limitations) from both papers, not just general themes.
2) Avoid simple combination or summary of contributions: Go further to find at least one of:
- Chain reasoning patterns (e.g., A→B from one work and B→C from the other suggests A→C).
- New research directions that neither paper alone proposes (methods, architectures, experimental setups).
- Surprising analogies, implicit principles, or shared structures not made explicit in either paper.
3) SELF-CONTAINED: Write the insight so a reader who hasn't read the papers can understand and act on it. Do not refer to Paper 1 or Paper 2.
Output format:
<think>Explain your reasoning for the insight. Step by step, show how specific aspects of Paper 1 and Paper 2 connect and why they lead to this insight.</think>

<insight>
A clear and self-contained statement of the insight (3-10 sentences).
</insight>

Figure 10: Prompt for insight generation. The summaries of two parent papers are in red.

C.1. Insight Generation Prompt

Below, the prompt utilized for the insight generation task is listed. The model is provided with the textual summaries of two distinct parent papers, which are highlighted in red in the accompanying figure. The prompt systematically instructs the model to analyze these summaries, synthesize their core methodologies or findings, and autonomously generate a novel scientific insight or proposed research direction that logically bridges and builds upon both foundational works.

C.2. Ancestor Identification Prompt

Here, we present the prompt designed for the ancestor identification task. The primary objective of this prompt is to instruct the model to determine the most influential parent papers for a given downstream target paper. By providing the model with the abstract or conceptual context of the downstream paper, the prompt asks the model to trace the scientific lineage backward and reliably identify which prior works from a candidate pool served as its direct conceptual ancestors.

C.3. Similarity Judge Prompt

Below, we outline the prompt used for automated evaluation and reward scoring via an LLM-as-a-judge framework. To quantitatively assess the quality and relevance of our generated insights, this prompt provides the evaluator model with both the human-verified ground-truth insight and the corresponding

You are an expert research analyst. Given this research paper, your task is to:
1. Identify two prior papers that this paper explicitly cites and builds upon by combining their ideas in a synergistic way.
  - The synergy should come from an insight that only emerges when information from both papers is synthesized.
  - It does not need to be the main insight or methodology of the paper. It could be a technique, approach, or design choice inspired by the two works — something that neither paper alone could fully support.
2. Explain your reasoning for why these two papers together provide the foundation for the insight.
3. Output the titles of the two papers in the following format:
<paper1>The first paper title...</paper1>
<paper2>The second paper title...</paper2>

Expected Output
- A brief explanation of the synergy (why combining these two papers matters).
- The titles of the two papers inside <paper1> and <paper2> tags.

Figure 11: Prompt for identifying two parent papers whose ideas are synergistically combined to produce the given paper's key insight, which is extracted as the ground-truth target y^*.

model-generated insight. The model is then instructed to critically compare the two texts and output a similarity assessment, evaluating them on semantic alignment, scientific coherence, and the degree of conceptual overlap.

C.4. Rewriting Insight Prompt

To ensure the generated insights could be evaluated consistently, we employed a rewriting step to standardize their format. Figure 13 displays the prompt used to instruct the model to rewrite the raw insight into a clear, standalone statement without losing its original semantic meaning.

```
Below is a research insight:
<research_insight>
{ground-truth insight}
</research_insight>
Below is a statement you need to evaluate:
<statement>
{model-generated insight}
</statement>
Task: Rate how similar the statement is to the research insight (1–10).

STRICT RULES:
- Similarity requires matching the SAME core idea.
- 'Inspired by', 'motivated by', or 'reasonable extension' ≠ same idea.
- Shared topic or keywords alone ≠ similarity.

Compare explicitly:
1) Key mechanism/method
2) Causal logic/workflow
3) Primary contribution/novelty

Downgrade if the statement:
- Omits the central mechanism
- Generalizes/abstracts the insight
- Proposes a new framework/direction

Scale:
1–2: Unrelated.
3–4: Shares topic but not the actual insight.
5–6: Partial conceptual overlap; misses at least one core mechanism or misaligns assumptions.
7–8: Strong match with only minor differences in mechanisms or assumptions.
9: Near-identical conceptual + causal + motivational mapping; only minor, non-substantive deviations.
10: Perfect: same ideas, same mechanism and roles, same objective/assumptions.
### Output Format
Format your response as follows:
<think>
Explain your reasoning for the rating you chose.
</think>
<rating>a number between 1 and 10</rating>
```

Figure 12: Prompt for assessing the similarity between the ground-truth insight and model-generated insight.

```
You are an expert research analyst.
You are given detailed summaries of two research papers:
<papers>
Paper 1:
{Paper 1 summary}

Paper 2:
{Paper 2 summary}
</papers>

You are also given an ORACLE RESPONSE that explains how a follow-up paper integrates the ideas from the two
papers above.

Your task:
- Reconstruct the reasoning process that could lead to the *same final insight* as in the ORACLE RESPONSE, but
using only the content of the two papers above.
- Do NOT mention, reference, or imply the existence of the follow-up paper, the oracle, or any external work.
- Your response must be self-contained and logically derived only from the two papers.
- Your response should preserve the full level of *conceptual and technical specificity* implied in the oracle
response.

<oracle_response>
{Raw synergy explanation}
</oracle_response>

OUTPUT FORMAT (return ONLY these three sections, nothing else):
<think>
Internal reasoning and planning for how to reconstruct the synergy between the two papers.
Explain how you will approach integrating their ideas and aligning the final insight with the oracle's conceptual
structure (without mentioning the oracle explicitly in the output itself).
</think>

<insight>
The final rewritten insight stated clearly and self-contained.
Assume the reader has not seen Paper 1 or Paper 2.
Do not refer to Paper 1 or Paper 2 by name or imply their existence.
The insight should capture the same underlying conceptual synergy and technical details as the oracle response.
</insight>

<reasoning>
Step-by-step explanation of how this insight is logically derived from the two papers.
Explicitly identify which core ideas or mechanisms from each paper combine to produce the final integrated
conclusion.
Be explicit and precise about the conceptual dependencies, theoretical links, and reasoning chain that lead to the
stated insight.
</reasoning>
```

Figure 13: Prompt for rewriting the insight as a standalone statement.

C.5. Hyperparameters for GIANTS-4B and Baselines

We detail the optimization and training configurations used for evaluating GIANTS-4B against the baselines below. The hyperparameter choices were guided by standard practices for aligning large language models, with slight adjustments made to accommodate our specific sequence lengths and batch sizes.

RL Hyperparameters. Table 4 outlines the configuration used during the reinforcement learning phase, specifically utilizing the GRPO algorithm. We applied a conservative KL penalty to prevent the model from drifting too far from the reference policy.

Hyperparameter	Value
Algorithm	GRPO [13]
Training steps	400
Train batch size	64
Group size	8
Max prompt length	3000
Max response length	1296
Learning rate	1×10^{-6}
Entropy coefficient	0.001
KL loss coefficient	0.001
KL loss type	`low_var_kl`
Sampling temperature (train / val)	0.6 / 0.6
Samples per prompt (train / val)	16 / 8
Max batched tokens	32768
Uneven Clipping (low / high)	0.2 / 0.5

Table 4: Key reinforcement learning hyperparameters used in our experiments.

Supervised Learning Hyperparameters. Table 5 details the setup for our supervised fine-tuning (SFT) baselines. Where multiple values were explored during our grid search, the final selected optimal configuration is marked in bold.

Hyperparameter	Value
Algorithm	SFT []
Epochs	10
Train batch size	64
Max length (input + output)	16384
Learning rate	5×10^{-5}, 1×10^{-5}, 5×10^{-6}, 1×10^{-6}
Weight decay	1×10^{-2}, 1×10^{-4}
Sampling temperature (train / val)	0.6 / 0.6
Learning rate schedule	Cosine annealing
Learning rate warmup ratio	0.01
Gradient checkpointing	True
Gradient accumulation steps	16

Table 5: Key supervised fine-tuning hyperparameters used in our experiments. Bold indicates the selected configuration when multiple values were explored.

D. Ablations with Diverse LM Judges

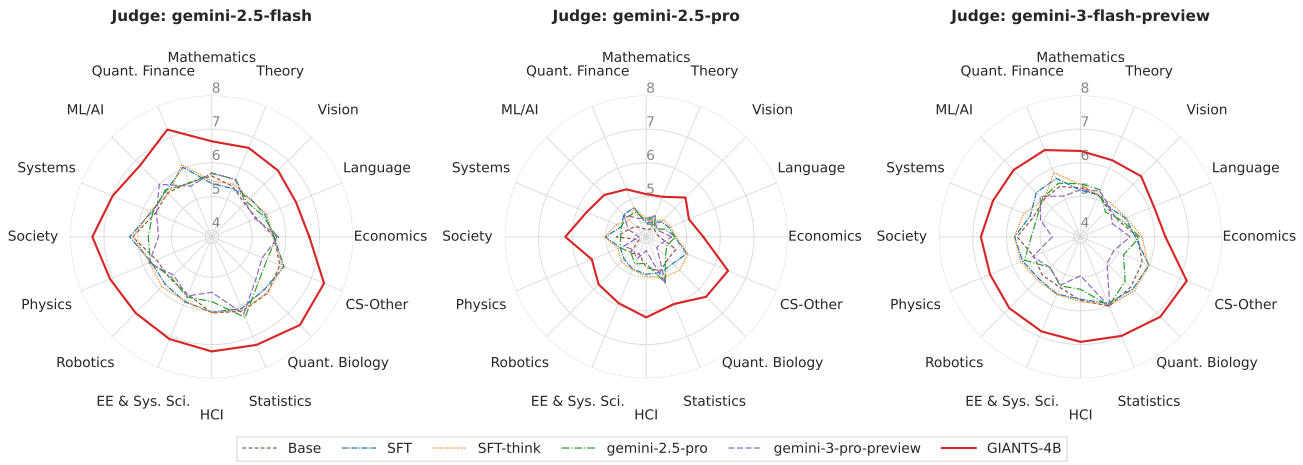

Figure 14: **Similarity scores on *Test-unseen-parents*.** This is the subset of GIANTSBENCH test set with only unseen parent papers.

E. More Human Eval Details

To validate the reliability of our automated metrics, we conducted a preliminary human evaluation study focusing on insight similarity. Annotators were tasked with scoring the semantic alignment between model-generated insights and ground-truth insights. To facilitate this process and ensure an unbiased environment, we developed a custom web interface, shown in Figure 16. Importantly, the human annotators were provided with the exact same scoring rubric and guidelines that were given to the LLM judge, allowing us to accurately measure the consistency between human judgment and our automated evaluation pipeline, with the exact context provided to the LLM judge.

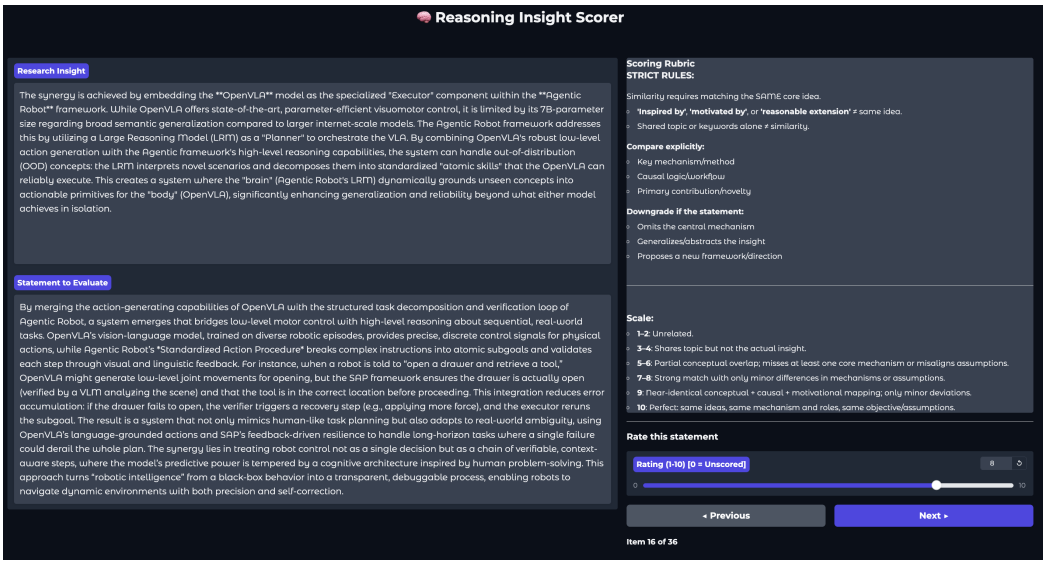

Figure 15: **Human Eval Labeling App** Here is the Gradio interface that was used by human labelers to measure the consistency of LLM similarity with human ratings. The rubric provided to the humans exactly matched that provided to the LLM judge for consistency (with markdown formatting for human interpretability).

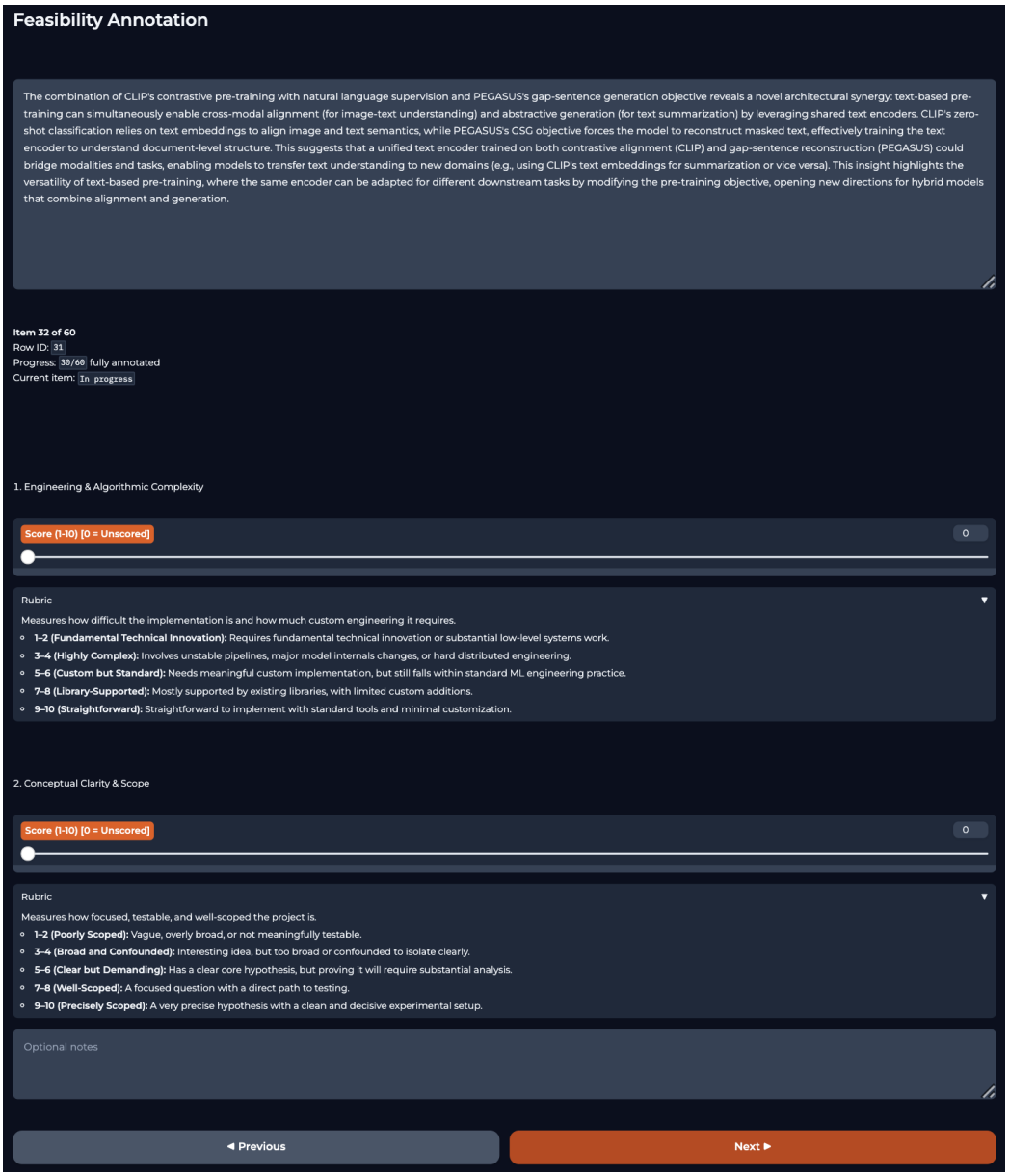

Figure 16: **Feasibility Human Eval Labeling App** Here is the Gradio interface that was used by human labelers to measure the feasibility of a research insight with human ratings. The rubric provided to the humans exactly matched that provided to the LLM judge for consistency (with markdown formatting for human interpretability).